twosuns

EVERY DEVIATION IN YOUR PLANT HAS A PRICE. WE SHOW YOU WHAT IT IS, EVERY DAY.

Cement plants do not lose their margin in major failures. They lose it in drift. A kiln running slightly heavy on heat. Free lime creeping past the band. A mill drawing more power per tonne than its curve allows. Each one is small enough to live with, and together they run into millions a year. On a 5,000 tonne per day line, kiln drift of 20 kcal per kg of clinker alone burns roughly **4,000 extra tonnes of fuel a year**. TwoSuns.ai runs on the operational data you already produce, flags these deviations against bands your engineers define, **attaches a cost per day to each one**, and records what your team decided to do about it.

1 INGEST

Connects to your historian, lab results, production and cost data. No rip and replace.

2 DETECT

Flags deviations against playbook bands that your own engineers define and own.

3 COST

Converts each deviation into money exposure per day, ranked by severity.

4 DECIDE

Recommends an action. The person accountable reviews it and dispositions it.

5 RECORD

Writes a permanent Decision Pack: who decided what, when, and on which data.

Every alert carries: Deviation vs playbook band · cost exposure per day · severity · the named owner · the disposition · full data lineage

WHERE IT BITES IN CEMENT

Kiln thermal regime and alternative fuels

Specific heat consumption, fuel rate drift and burning zone excursions, monitored by kiln playbooks running in production today. The same bands govern alternative fuels, holding substitution rate against thermal stability and cost per gigacalorie.

Clinker quality and free lime

Free lime, lime saturation and burnability drift are caught as costed deviations. Off spec clinker is priced at the moment it is made, not discovered later at the cement mill.

Raw mix and grinding energy

Raw meal chemistry stability, mill power per tonne and separator efficiency. Electrical energy is the quiet margin killer, so kWh per tonne is held to its band and ranked in money.

Product economics and allocation

A per product cost engine across the full slate, covering ordinary, sulphate resistant and oil well cements in bulk and packed form, with pricing position and allocation pressure visible in month rather than at close.

Carbon exposure and energy intensity

CO_2 per tonne of clinker is becoming a direct cost line through carbon pricing, border adjustments and lender ESG terms. An audit grade decision trail is the difference between a carbon number and a carbon dispute.

WHAT THE PLATFORM DELIVERS

Costed alerts, not dashboards

Every deviation arrives as a money figure with severity and trend, engineered for the question "what is this costing us today?"

Playbooks your engineers own

Detection logic lives in plain-text playbooks, readable bands and rules, not buried model weights. Site admins set defaults; operators apply governed overrides.

Audit-ready Decision Packs

Every material decision, including stoppage and restart calls, is captured with its data, the alternatives considered and its owner. Built for incident inquiries, compliance reviews and board reporting.

A financials layer the CFO can trust

Per product cost, pricing and allocation, with every figure tagged as measured, GL derived or modelled. Finance and operations then argue about the decision rather than about the data.

A financials layer the CFO can trust

Per-product cost, pricing, and allocation, every figure confidence-tagged *measured / GL-derived / modelled*, so finance and ops argue about the decision, not the data.

Proven on cement kilns. TwoSuns runs today on cement kiln and drilling operations in the Gulf region, live kiln monitoring with operator-governed thresholds, against live production data.